Outline

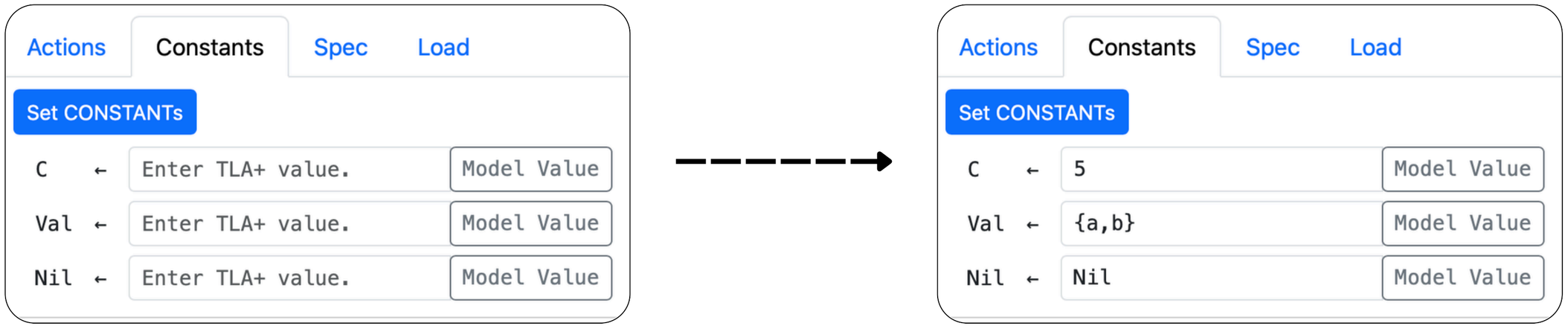
1. How to Run the Animation?

2. Expalnation of the Animation.

How to run Animation?

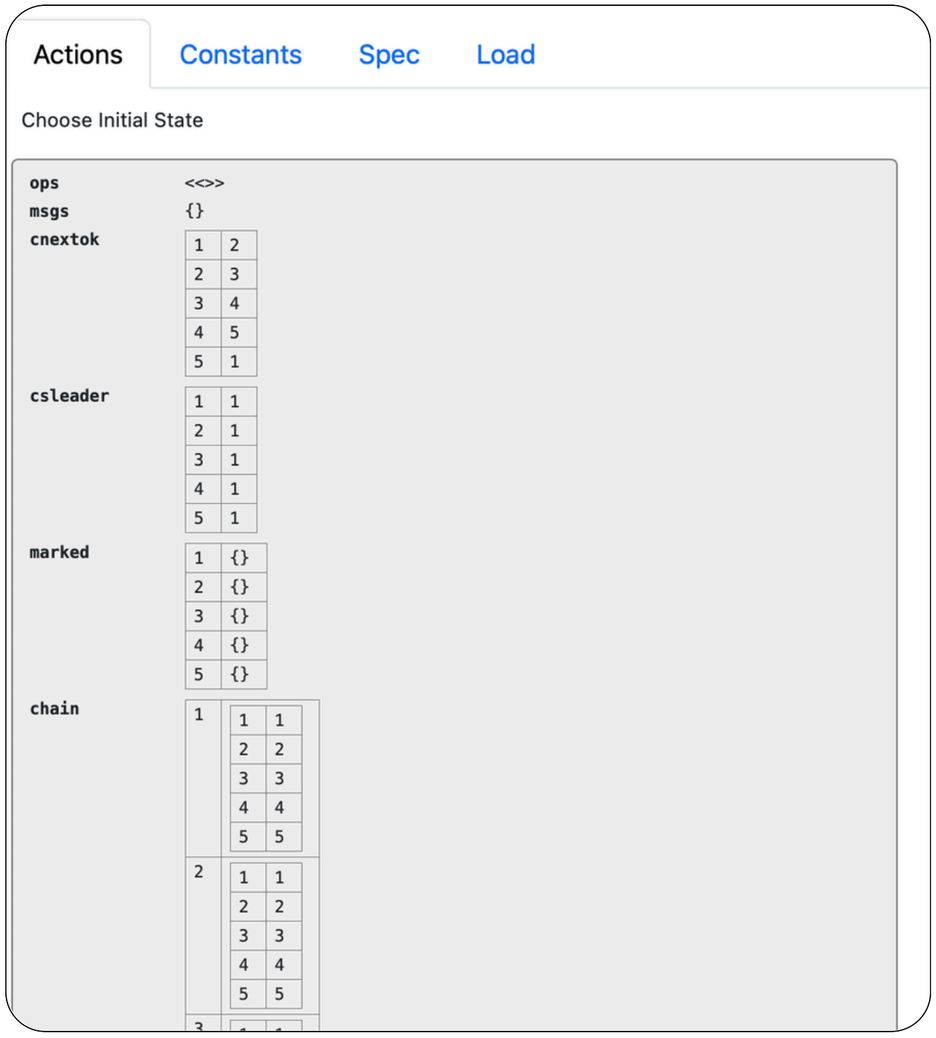
From the below link you can run the animation.
https://chainraft.web.app

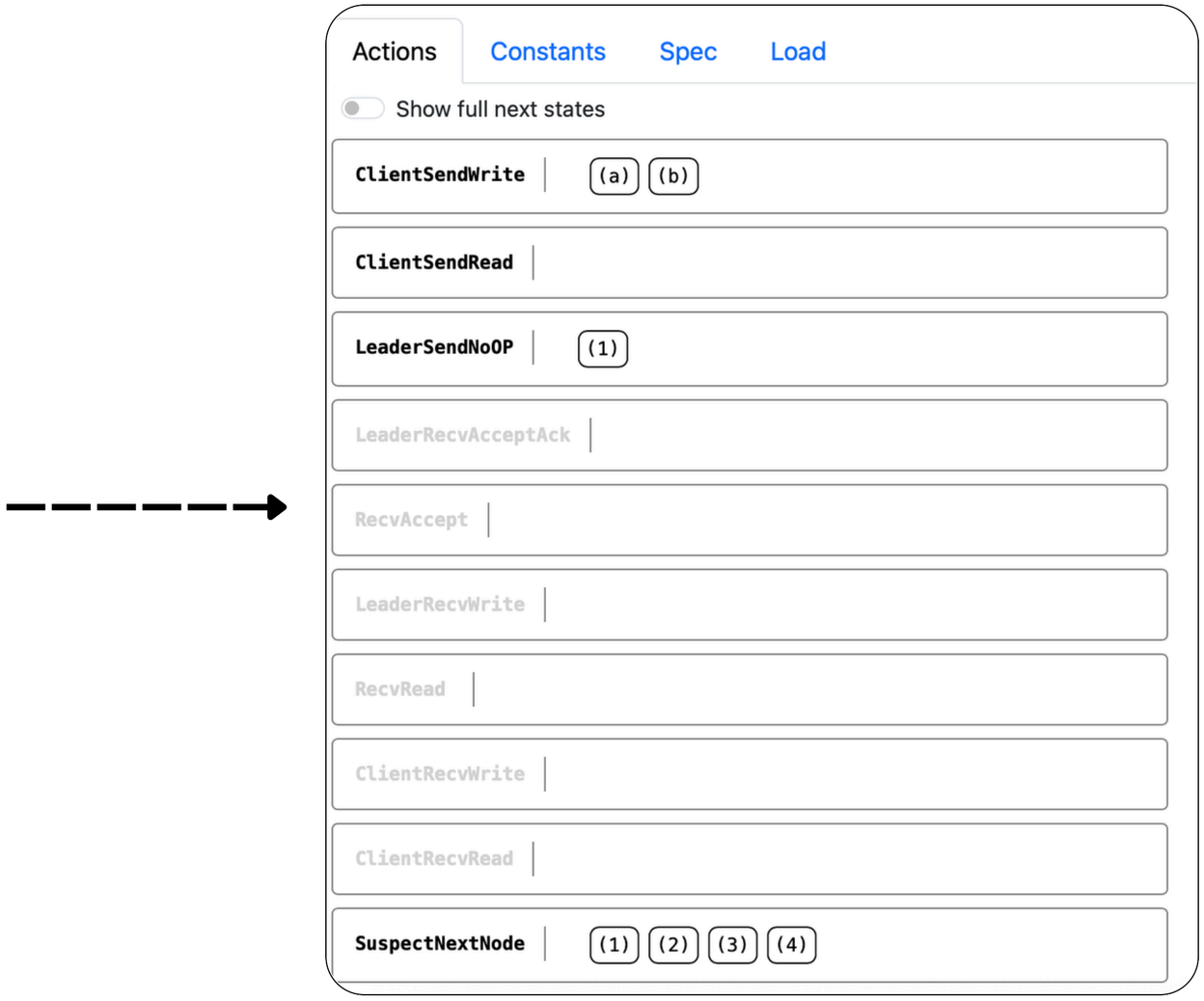
Step 1: Write the Constants and Click on Set CONSTANTs



How to run Animation?

Step 2: Click on the Initial State & you'll see all the actions possible from this state in the Actions tab.



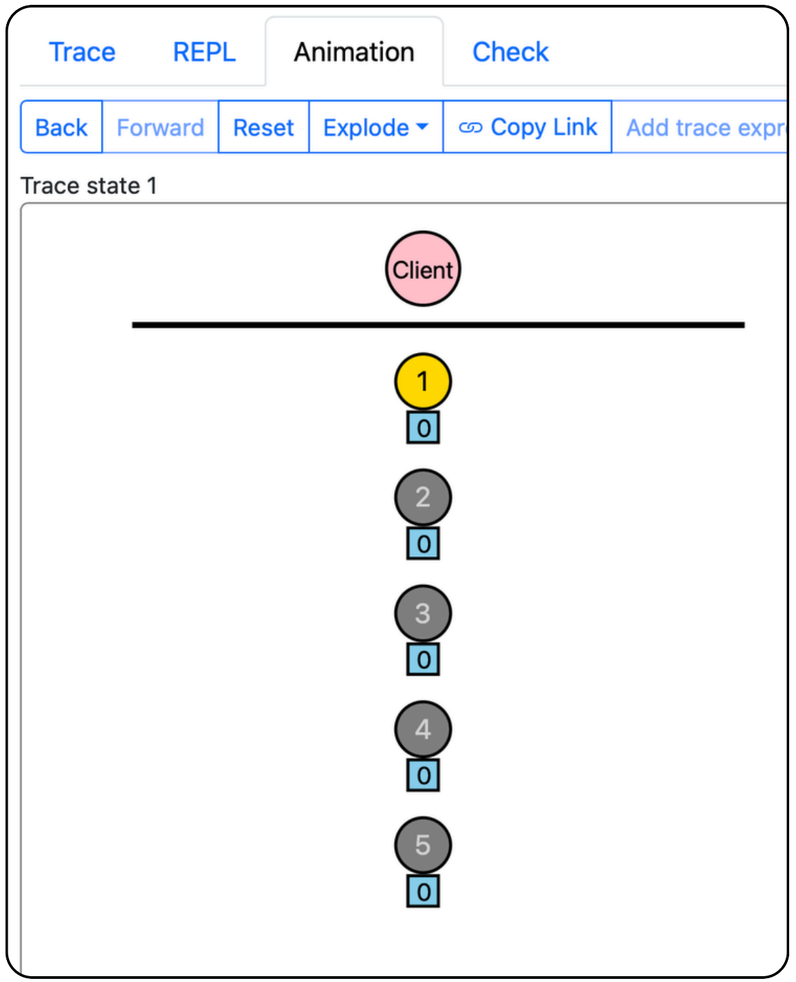


How to run Animation?

Step 3: On the right hand side you can see the tabs. To see the trace of the behaviour click on the Trace tab and to see the animation click on the Animation tab.



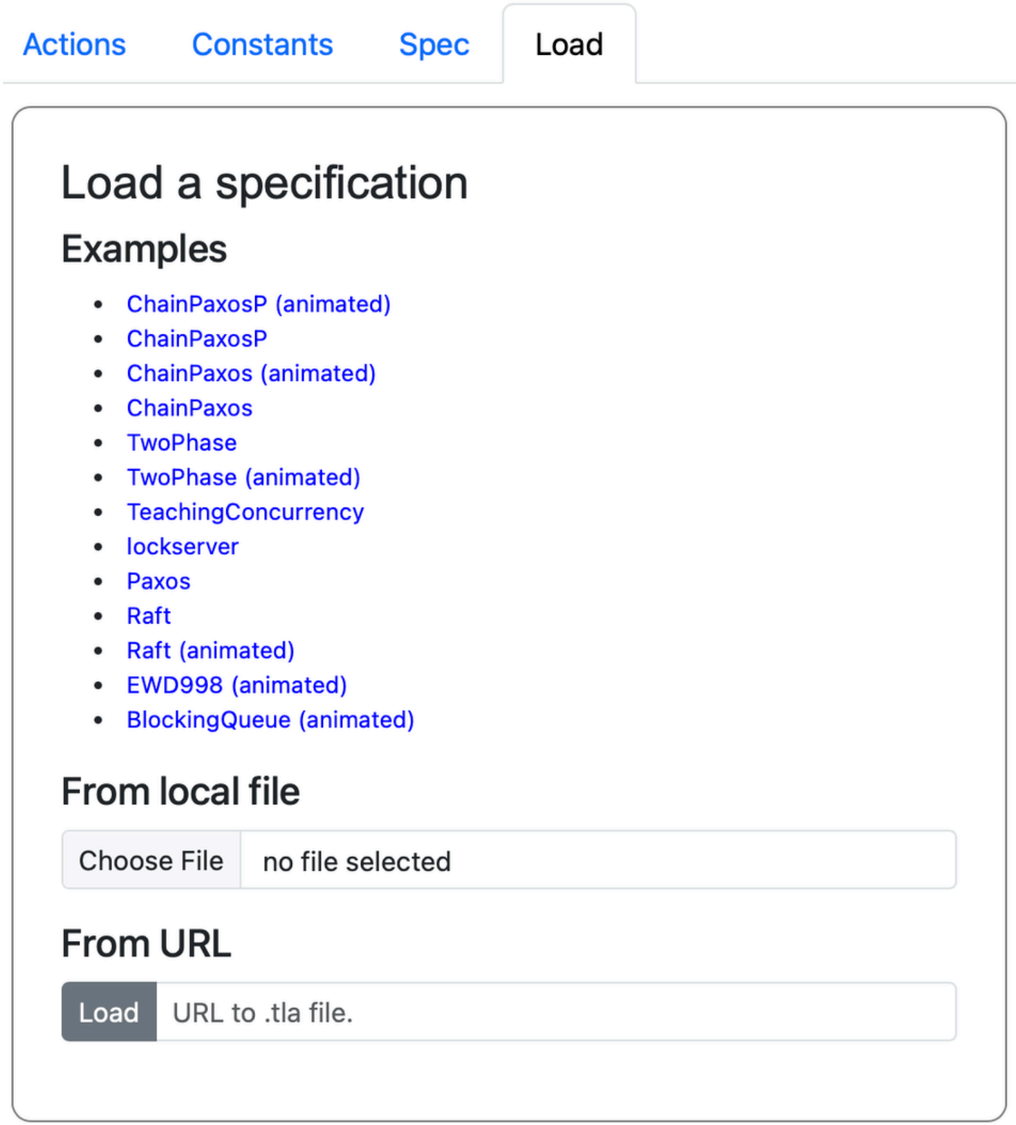




How to run Animation?

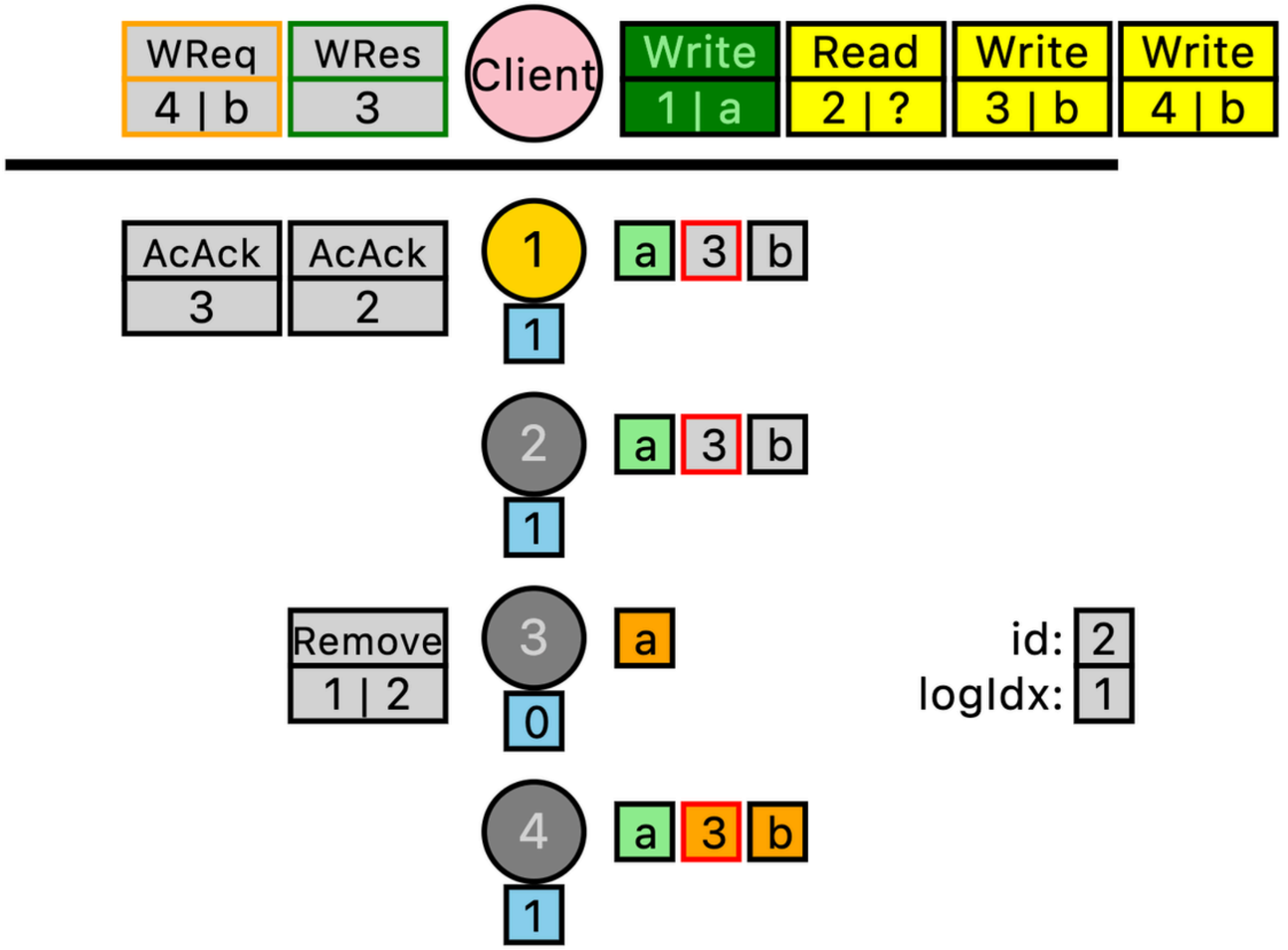
Step 4: Now by choosing different actions you can see the animation going.

You can also Load other specs from the Load tab by clicking on them & run them using the same steps.

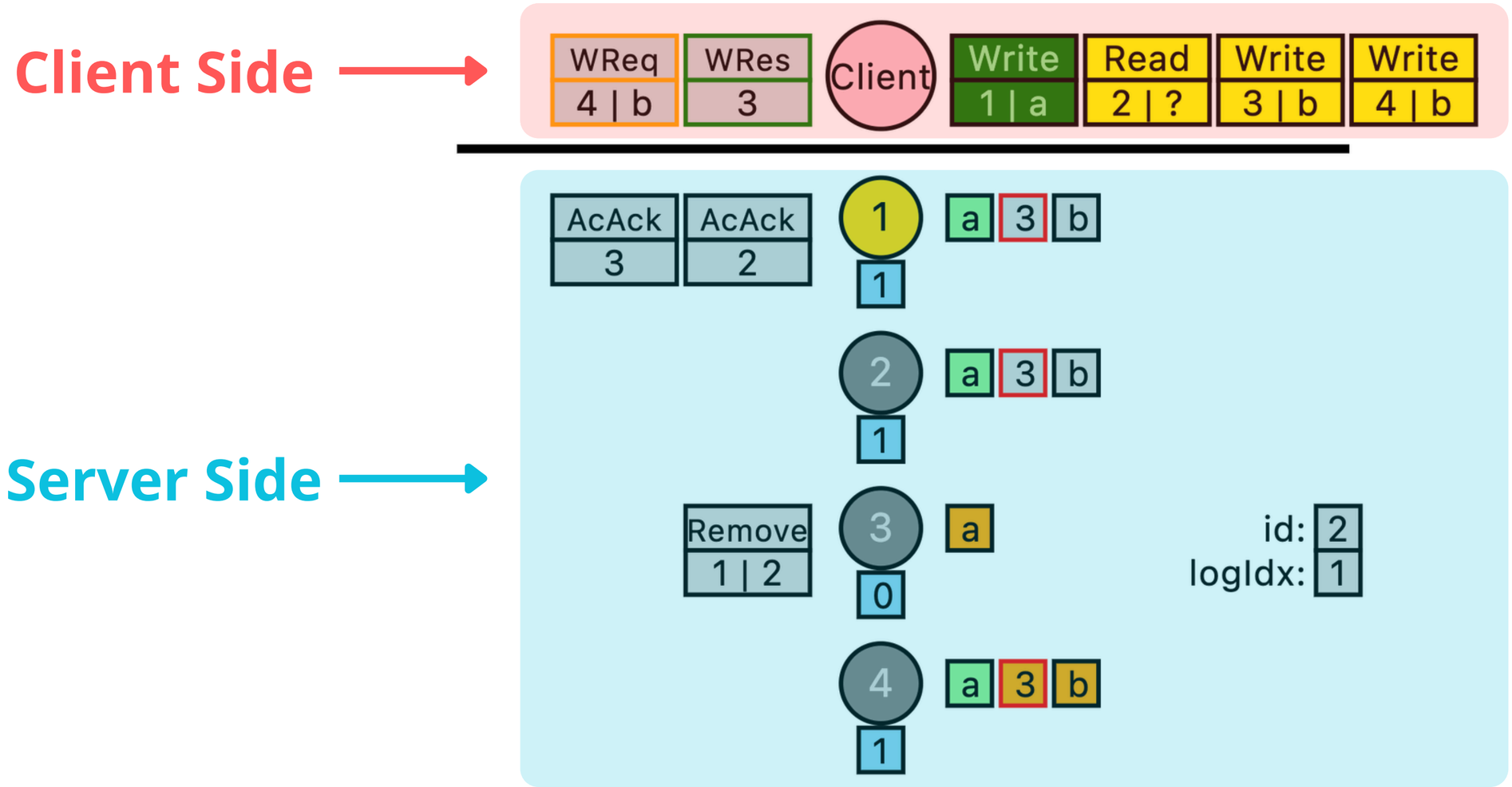


Exaplantion:

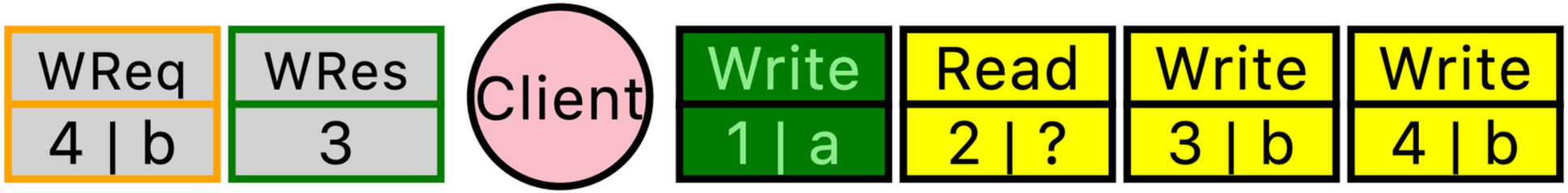
Overall Animation



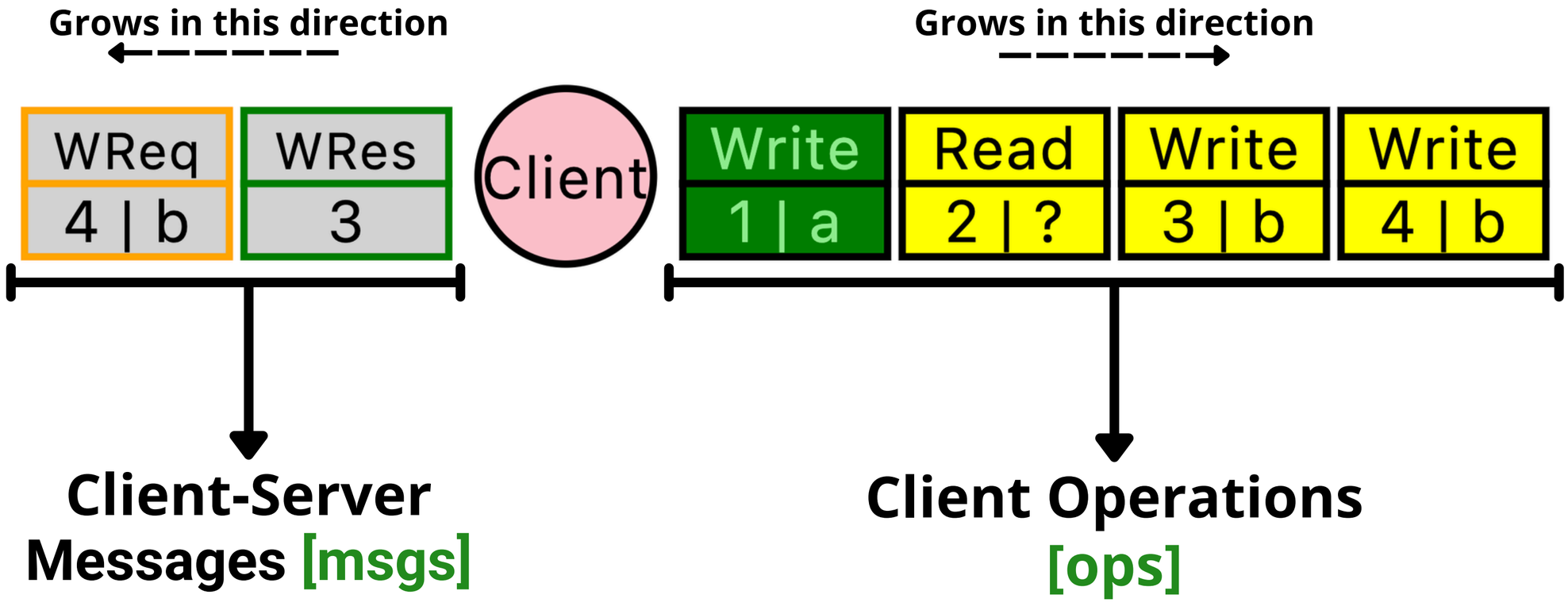
Overall Animation



Client Side



Client Side


Grows in this direction
Grows in this direction
WReq
4 | b
WRes
3
Client
Write
1 | a
Read
2 | ?
Write
3 | b
Write
4 | b
Client-Server
Messages [msgs]
Client Operations
[ops]

Client Side

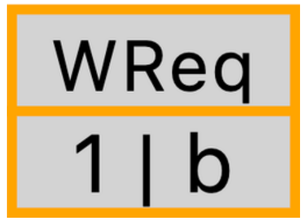
Client-Server Messages: msgs

Message Type

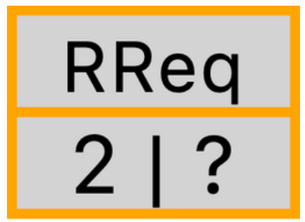
Border Color	Type
	Request message
	Response message
	RemoveNode
	AddNode

Text	Type
WReq \ WRes	Write Req \ Res
RReq \ RRes	Read Req \ Res
Remove	RemoveNode
Add	AddNode

Client Request



Write Request with value = b for operation id = 1

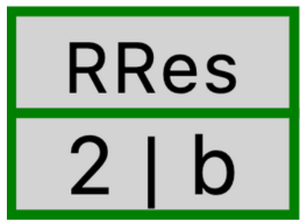


Read Request for operation id = 2

Response



Write Response for operation id = 1



Read Response with value = b for operation id = 2

Client Side

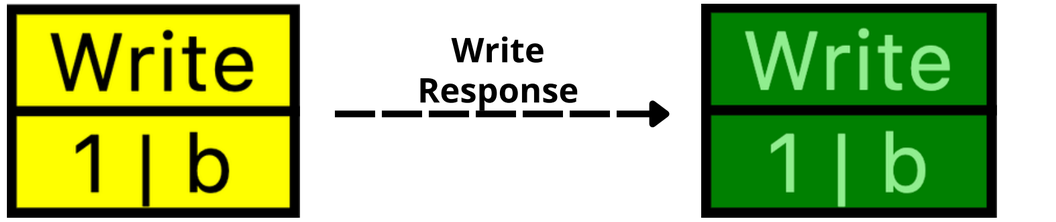
Client Operations : ops

Operation Type & status

Text	Type
Write	Write Operation
Read	Read Operation

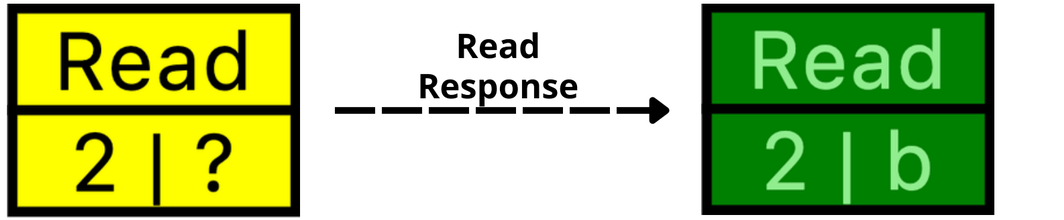
Filled Color	Status
	Pending
	Completed

Transition



Pending Write Operation with value = b & id = 1

Completed Write Operation with id = 1

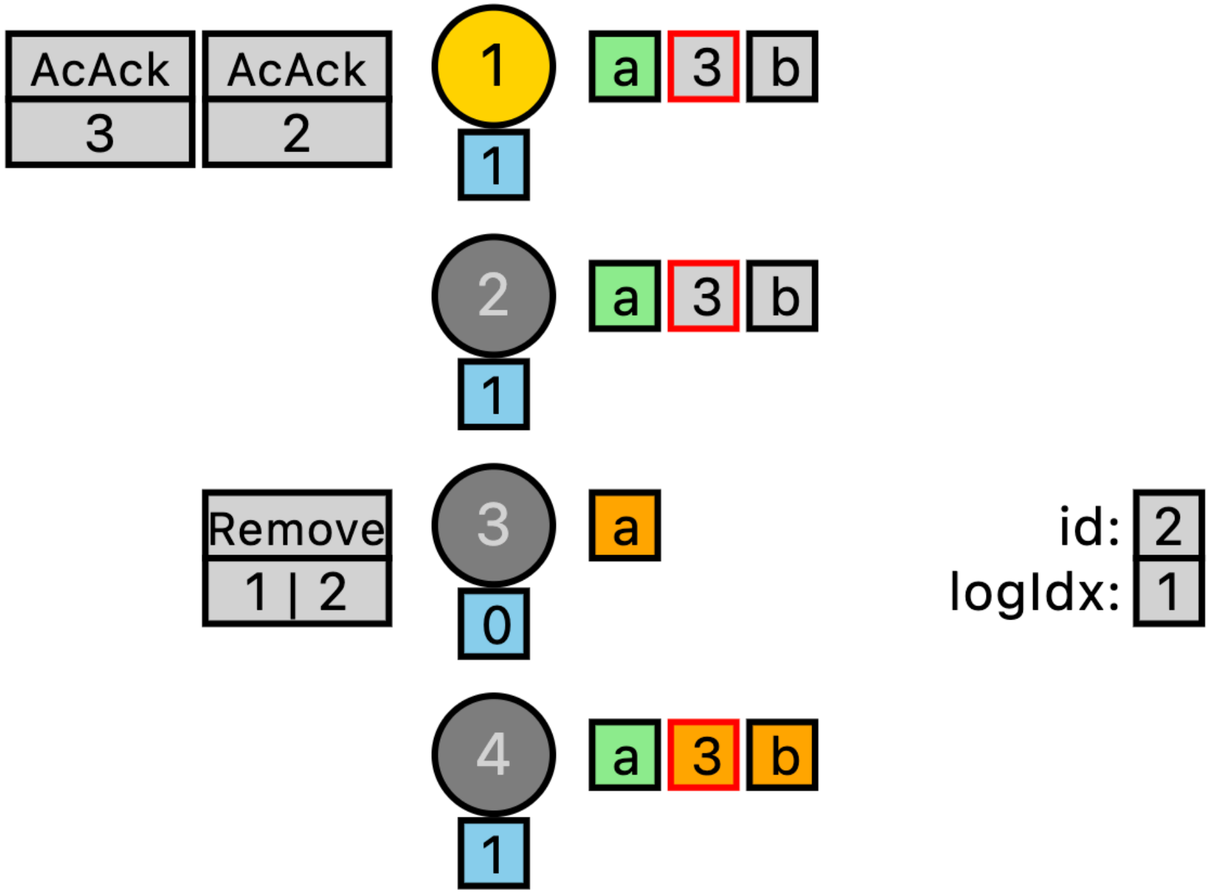


Read Request for operation id = 2

Read Response with value = b for operation id = 2

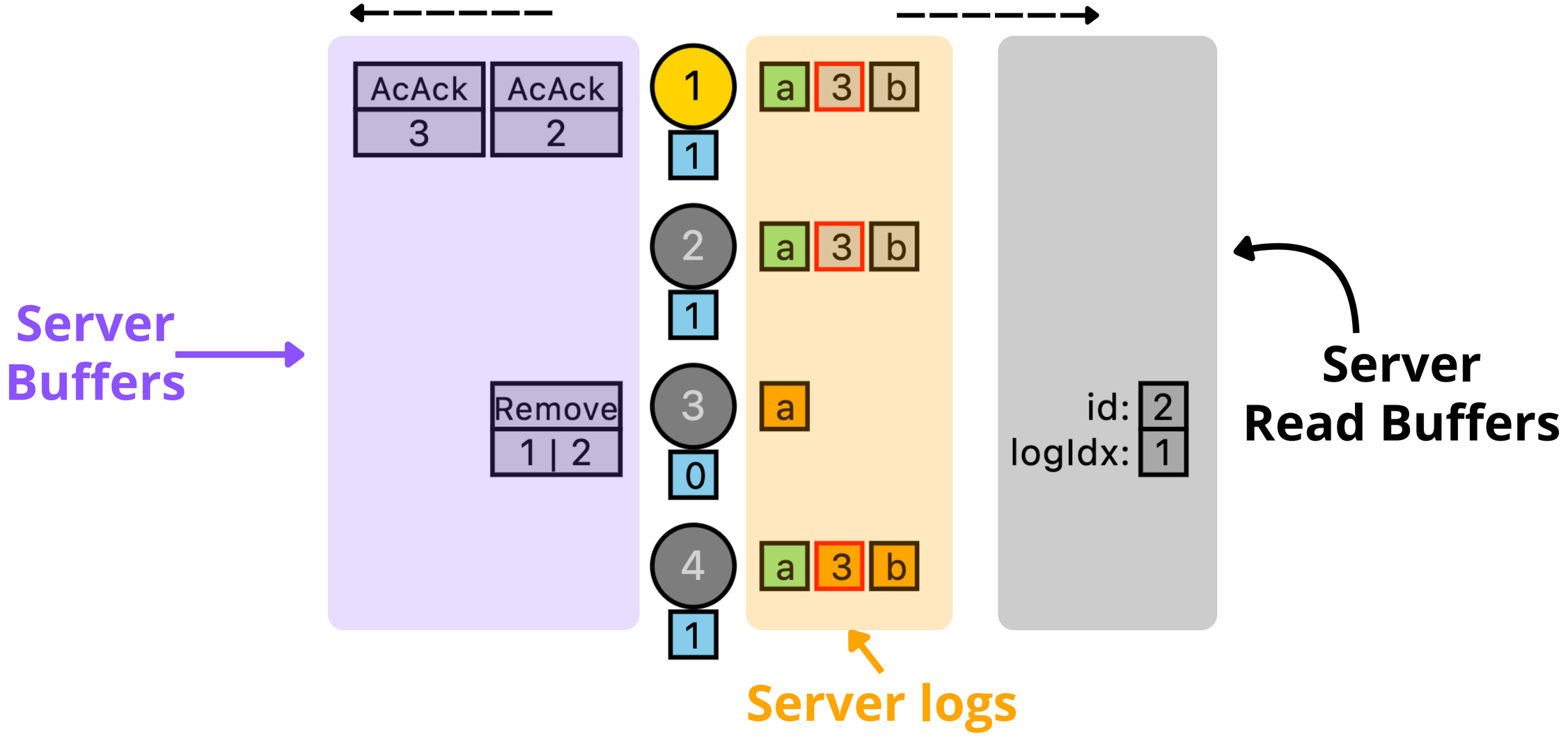
Server Side

All Server view



Server Side

All Server view

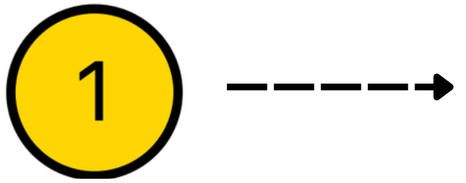


Server Side

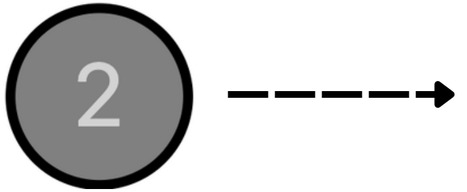
Nodes

Node Status

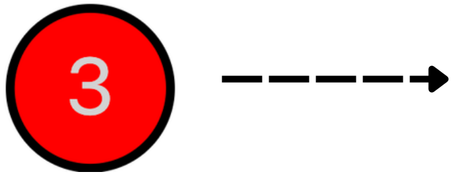
Filled Color	Status
	Leader (Active)
	Non-Leader (Active)
	Removed Node (Inactive)



Node-1 is Leader & active



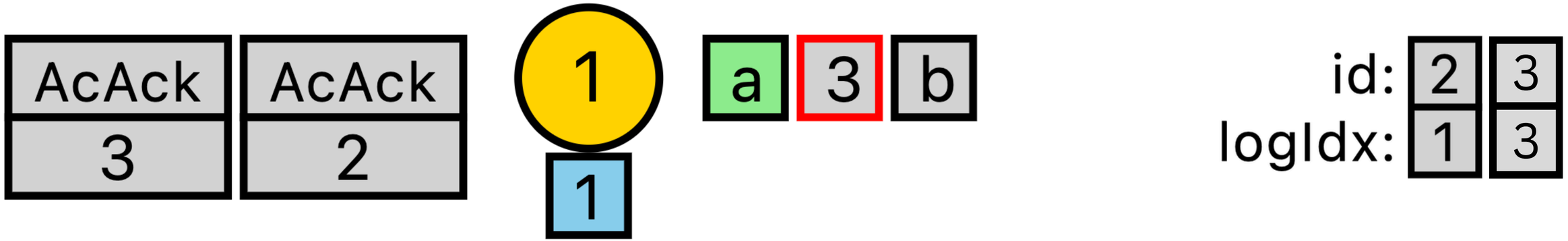
Node-2 is active non-leader in the Chain



Node-3 is **Removed** from the Chain

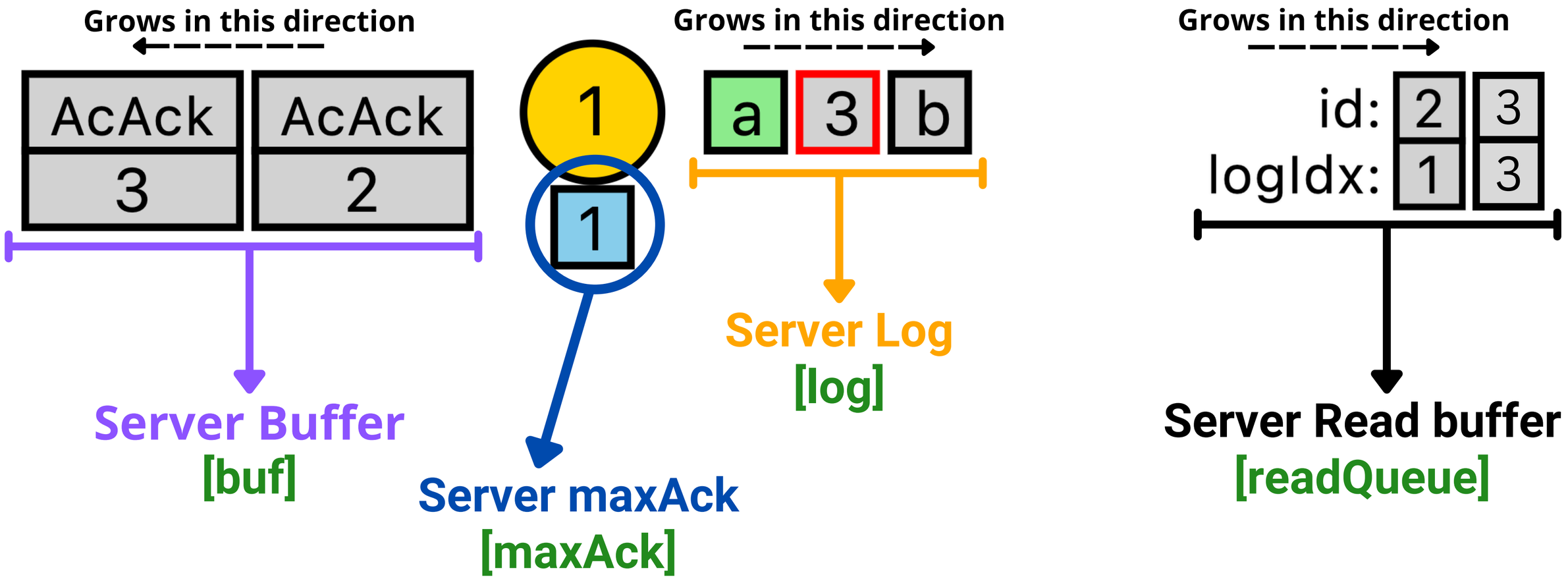
Server Side

Single Server view



Server Side

Single Server view

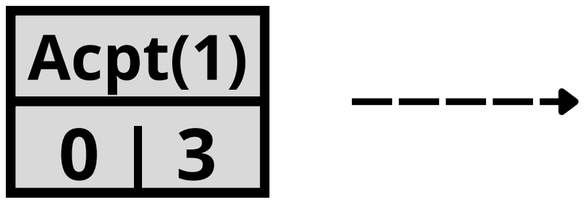


Server Side

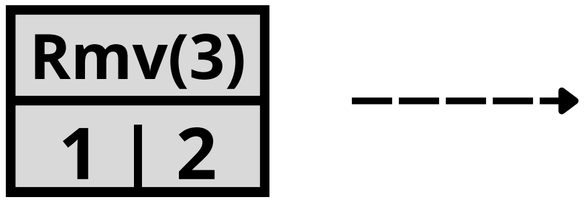
Server Buffer: buf

Message Type

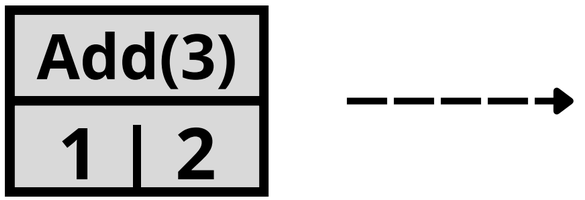
Text	Type
Acpt(_)	Accept With Write/NoOP Operation
Rmv(s)	Accept With RemoveNode(s) Operation
Add(s)	Accept With AddNode(s) Operation
AcAck	Accept Ack message



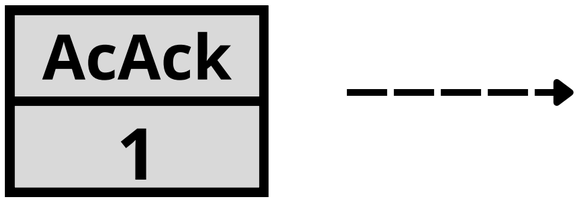
Accept message with
idx = 1
maxAck = 0
#Accepts = 3



RemoveNode for
s = 3 with
maxAck = 1
#Accepts = 2



AddNode for
s = 3 with
maxAck = 1
#Accepts = 2



Accept Ack for
idx = 1

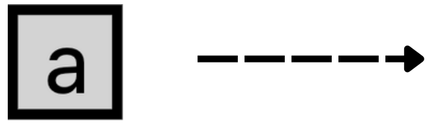
Server Side

Server Log: log

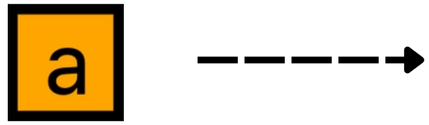
Log Entry Type & Status

Border Color	Type
●	Written Value
●	RemoveNode
●	AddNode

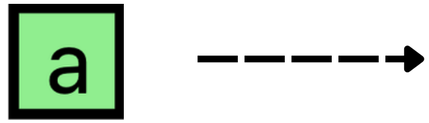
Filled Color	Status
○	Not decided Not Acked
●	Decided Not Acked
●	Decided Acked



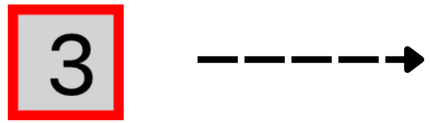
Value a is neither decide nor Acked



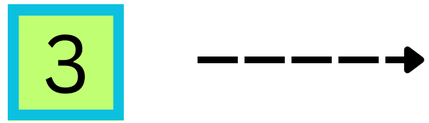
Value a is decided but not Acked



Value a is Acked



RemoveNode 3 neither decided nor Acked



AddNode 3 is Acked

Server Side

Server Read Buffer: readQueue

Term Meanings

Term	Meaning
id:	id of the read operation waiting in the read buffer
logIdx:	The maxAck for which it's corrosponding read is waiting

